

Vendor Performance Data & Business Analysis

A sequenced review of data quality, exploratory patterns, supplier economics, inventory exposure, and management-action design.

Decision Statement. The portfolio delivers **\$452.06M in sales** and **\$138.47M estimated gross profit** at a **30.66%** weighted margin. Procurement is heavily concentrated: the top ten vendors represent **65.33%** of total spend. Year-end inventory stands at **\$79.70M at retail**, making supplier continuity, inventory-value reduction, and controlled promotions the immediate management priorities.

\$452.06M TOTAL SALES Fiscal 2024	\$321.90M PROCUREMENT SPEND Received POs	\$138.47M EST. GROSS PROFIT Cost-matched
30.66% WEIGHTED MARGIN Revenue-weighted	65.33% TOP-10 CONCENTRATION Share of spend	\$79.70M ENDING INVENTORY At retail value

Executive Recommendations

- Concentration risk:** establish continuity plans and alternate-source documentation for the ten vendors driving the first 65% of procurement spend.
- Inventory action:** review replenishment and markdown options for high-retail-value, low-sell-through brands before the next ordering cycle.
- Promotion pilot:** test a controlled subset of the 485 low-sales, high-margin candidates with minimum-margin guardrails.
- Scorecard:** track weighted margin, sell-through rate, ending inventory value, and supplier concentration as a monthly leadership scorecard.

§ 1. Analytical Sequence and Data Foundation

The case study follows a deliberate **Data Analyst** → **Business Analyst** sequence. Exploration establishes data shape and reliability first; decision analysis then converts those findings into commercially actionable outputs.

Stage	Analytical Purpose	Primary Output
Understand	Confirm source roles and establish a safe analytical grain	Source map and join controls
Explore	Describe distributions, trends, outliers, and relationships	EDA charts and diagnostic tables
Measure	Define weighted, decision-safe KPIs	Reconciled KPI scorecard
Diagnose	Locate concentration, promotion, and inventory risk	Prioritised vendor and brand lists
Recommend	Connect evidence to owners, timing, and success measures	30/60/90-day action plan

Source-System Scale

Source	Approx. Rows	Analytical Role
Sales	12.8 M	Revenue, units, vendor-brand movement, monthly trend
Purchases	2.37 M	Procurement quantity, spend, and weighted cost
Beginning inventory	204 K	Opening stock availability
Ending inventory	221 K	Closing stock and retail-value exposure
Vendor invoices	5.3 K	Freight and invoice context
Purchase prices	12.2 K	Cost and price reference

Grain and Analytical Controls

Sales and purchases are aggregated independently to one **vendor-brand** row before joining. This design prevents changing purchase prices from duplicating sales records. The analytical layer contains **11,272 unique vendor-brand rows**; purchase cost is available for **99.90%** of sales. Profitability calculations exclude the unmatched 0.10% rather than assigning a zero cost.

Inventory reconciliation check. 4,219,275 beginning units + 33,584,377 purchased units – 32,917,876 sold units = 4,885,776 ending units. **Portfolio-level residual is exactly zero.**

§ 2. Exploratory Analysis — Distributions

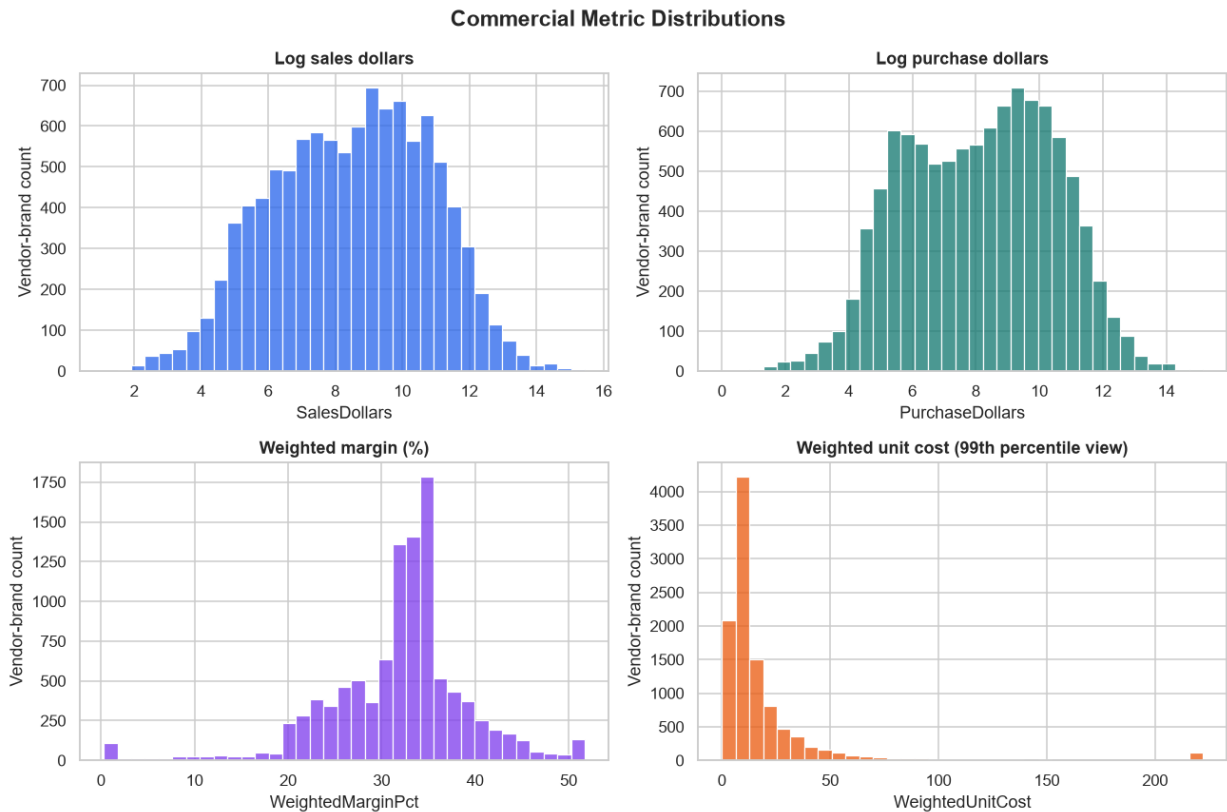


Figure 1: Distribution profiles across vendor-brand observations (log scale; right-skew evident in sales and procurement values).

Metric	Median	Mean	95th Pct.	Maximum
Sales dollars	\$5,599.72	\$42,959.43	\$175,261.91	\$5.10M
Purchase dollars	\$3,835.15	\$30,586.65	\$120,565.16	\$3.81M
Weighted unit cost	\$10.45	\$23.41	\$61.60	\$4,264.70
Weighted margin	32.89%	31.74%	43.26%	100.00%

What the Distributions Mean

Sales and procurement values are **strongly right-skewed**: a small number of vendor-brand combinations are substantially larger than the typical observation. Medians and percentile bands therefore give a more representative description than means alone. Extreme values are retained because commercially dominant vendors and brands are not automatically data errors.

Outlier Profile

Metric	IQR-Flagged Rows	Share of Analytical Rows
Sales dollars	1,350	12.84%
Purchase dollars	1,334	12.69%
Weighted unit cost	1,056	10.04%
Weighted margin	608	5.78%

IQR flags are review signals, not deletion rules. Removing flagged rows would erase many of the suppliers and brands most relevant to portfolio-level decisions.

§ 3. Exploratory Analysis — Monthly Trend

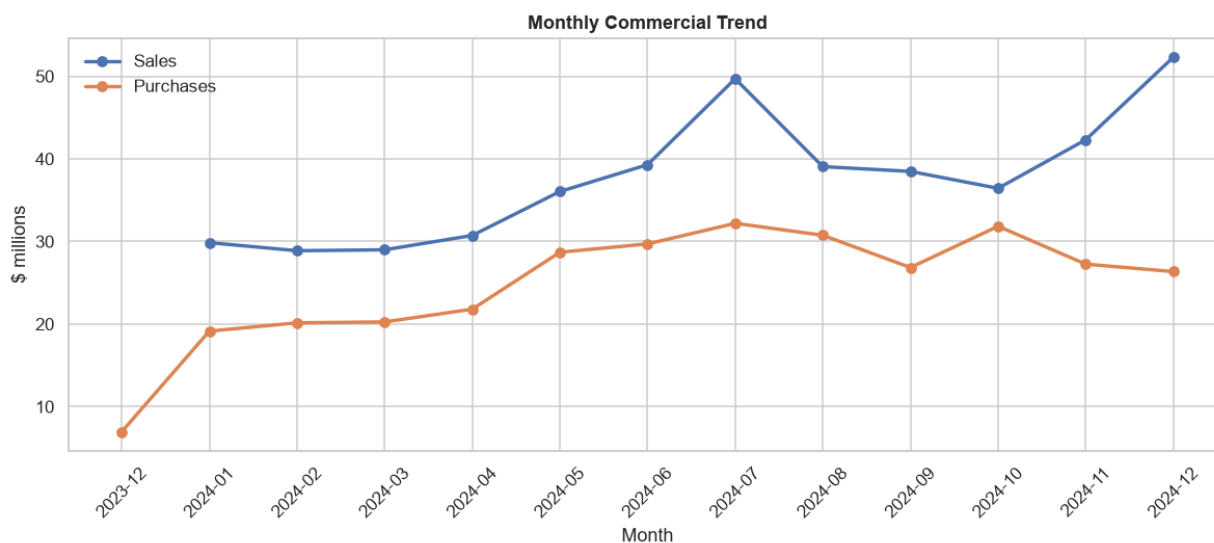


Figure 2: Monthly sales and procurement values illustrating seasonal build and year-end peak.

Observed Pattern

Sales rise from roughly \$29M per month in the first quarter to **\$49.70M in July** and a year-high **\$52.31M in December**. Procurement peaks earlier — around July and October — consistent with inventory positioning ahead of stronger sales periods. The December 2023 purchase value is retained as a timing observation and is not paired with 2024 sales figures.

Decision use. Category managers should compare purchasing lead times with the July and year-end sales peaks. The trend supports timing analysis but does not by itself prove supplier delay, over-ordering, or promotion lift. Those claims require additional investigation.

Transaction Diagnostics

Source	Rows	Missing Key Rows	Non-Positive \$
Sales	12,825,363	0	55 (0.0004%)
Purchases	2,372,474	0	153 (0.0064%)

The small number of non-positive values is visible and retained for business-rule review. Key-field completeness is sufficient for the vendor-brand analysis.

§ 4. Exploratory Analysis — Vendor and Brand Landscape

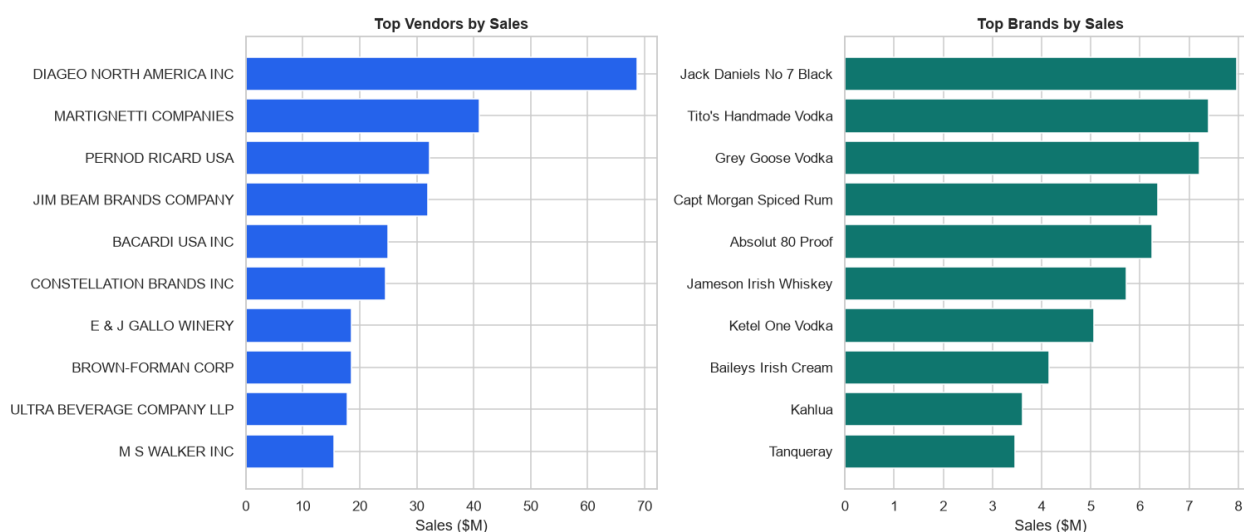


Figure 3: Top vendors and brands by sales; high sales alone does not confirm attractive margin, low supplier dependency, or healthy inventory turnover.

The sales leaders establish portfolio context before performance is judged. High sales volume alone is insufficient evidence of attractive margin, low supplier dependency, or healthy stock movement. This is why the Business Analyst layer combines sales with weighted cost, procurement share, and inventory measures.

Relationship Analysis

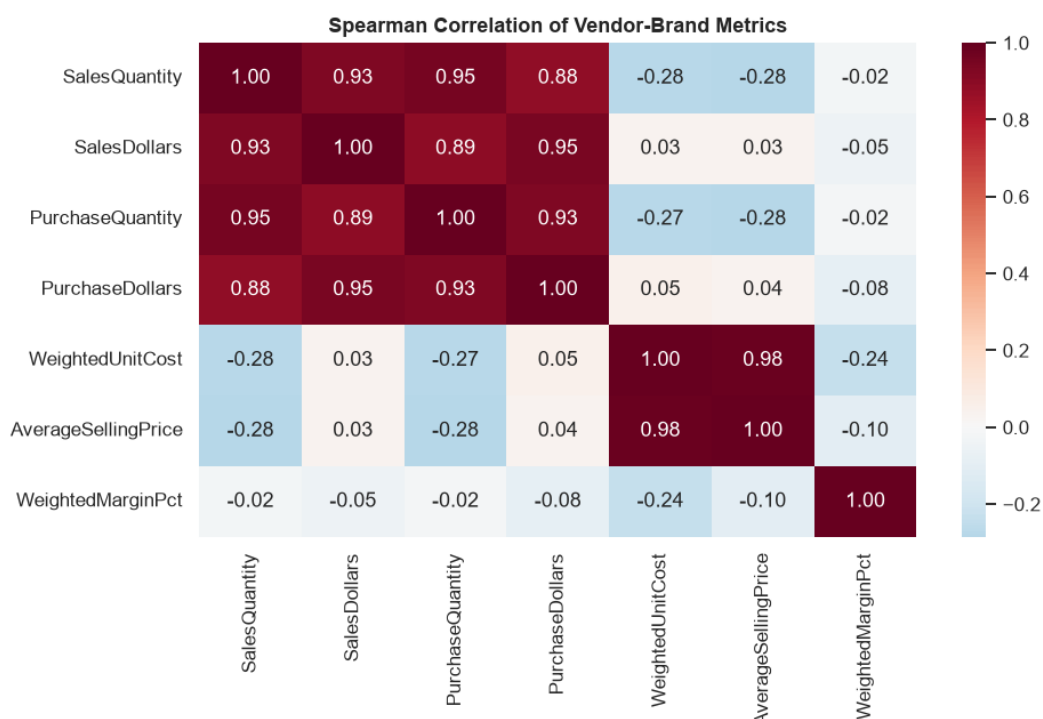


Figure 4: Spearman rank correlation matrix across key commercial metrics.

Spearman correlation is used because commercial measures are right-skewed. Sales and procurement scale move together strongly (expected given shared volume drivers), while unit cost and margin relationships are weaker and shaped by product mix. All associations are treated as descriptive, not causal.

§ 5. Business Analysis — Supplier Concentration

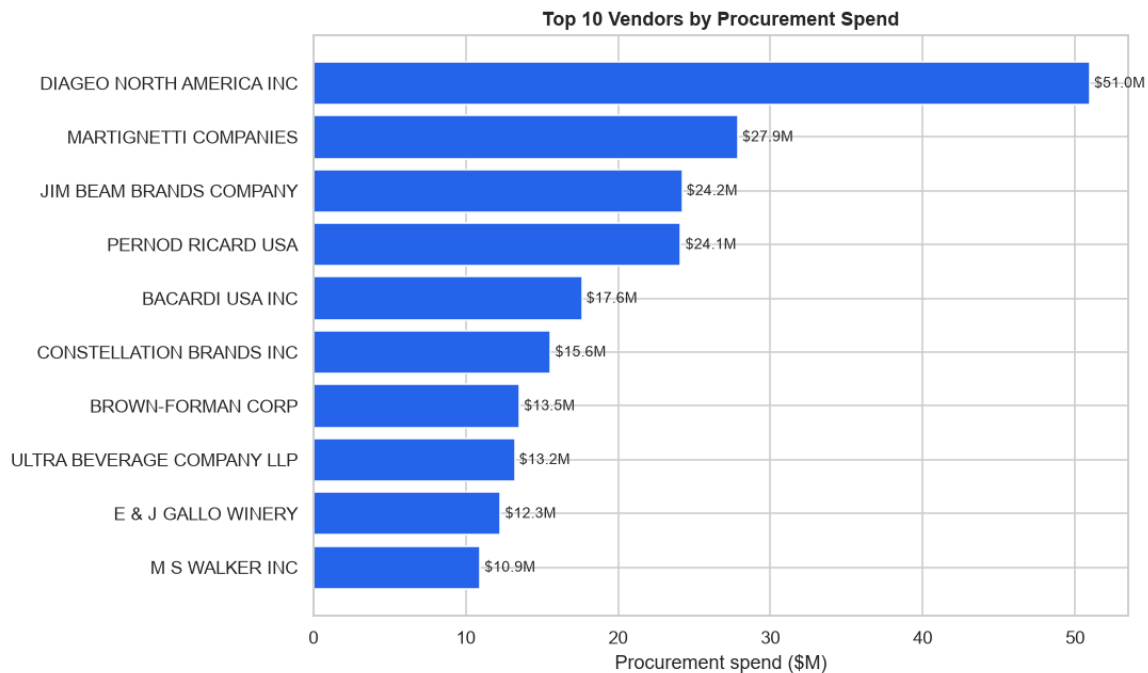


Figure 5: Procurement spend concentration: cumulative share by vendor rank.

The top ten vendors account for **65.33% of procurement spend**. DIAGEO NORTH AMERICA INC alone represents \$50.96M — **15.83%** of total procurement spend. Concentration is not automatically poor performance, but it materially increases the financial and service impact of any supply disruption.

Management Response

Owner	Required Action	Success Measure
Procurement	Document alternate sources, lead times, contract coverage, and recovery options for the top ten vendors.	Continuity coverage recorded; concentration monitored monthly.
Finance	Track freight once per vendor as a share of spend; do not allocate vendor totals across all brands.	Freight-to-spend variance explained without duplicating totals.
Analytics	Monitor spend share and cumulative concentration; escalate material shifts.	Changes above defined threshold escalated with evidence.

Decision principle. Manage supplier concentration alongside margin dollars and service risk. A vendor that dominates spend is not automatically underperforming — risk-adjusted assessment requires continuity and margin data together.

§ 6. Business Analysis — Brand Opportunity and Inventory Risk



Figure 6: Low-sales, high-margin brands as a controlled promotion test pool.

The screen identifies **485 promotion candidates** sitting in the bottom sales quartile and top weighted-margin quartile. This is a prioritised starting list, not a recommendation to discount every candidate. Piloting 10–15 brands with explicit margin guardrails is the appropriate first step.

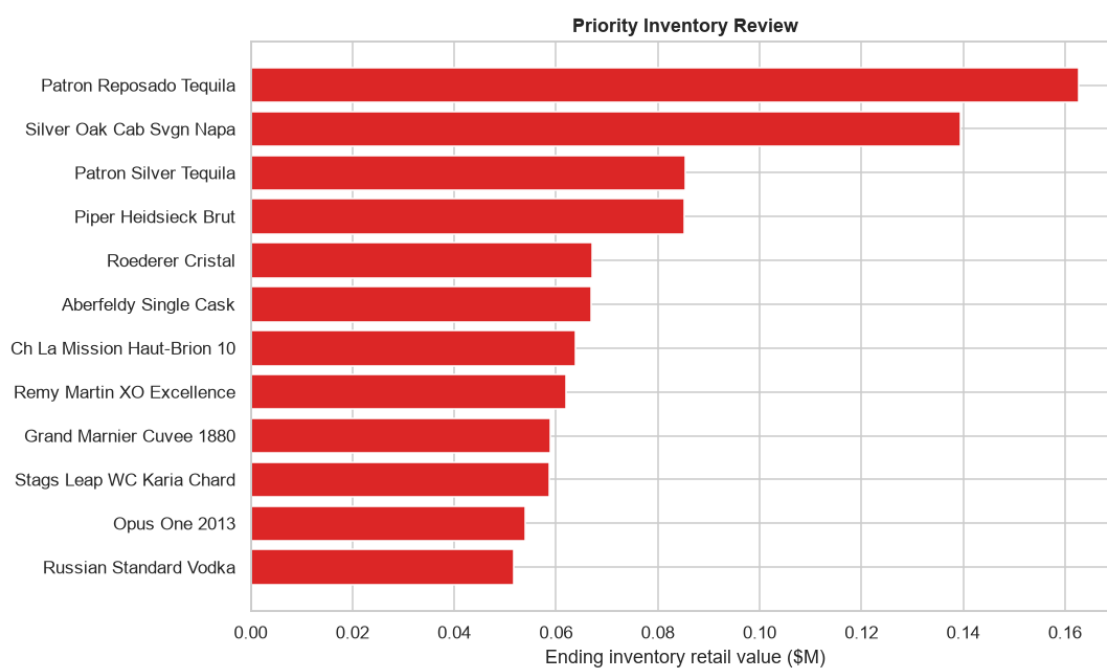


Figure 7: Priority inventory: high ending retail value combined with weak sell-through rate.

Ending inventory totals **\$79.70M at retail**. Management should begin with replenishment review, supplier-return options, and measured markdown tests for high-value, low-sell-through brands. Inventory snapshots do not carry a vendor key, so exposure is assessed at the brand level.

Inventory is valued at retail, not cost. The \$79.70M figure overstates cash-at-risk and should not be compared directly with procurement spend without a cost-to-retail conversion.

§ 7. Statistical Assessment and Interpretation

Purchase Volume vs. Unit Cost

Median unit cost declines from \$13.60 in the lowest purchase-volume quartile to \$8.20 in the highest quartile. This pattern is **descriptive, not proof of a bulk discount**: the quartiles contain different products, sizes, and price points. A causal claim would require matched products with price variation at different order quantities, or a controlled purchasing experiment.

Volume Quartile	Vendor-Brand Obs.	Median Unit Cost	Median Margin
Q1 Low	2,629	\$13.60	32.89%
Q2	2,628	\$11.03	33.32%
Q3	2,628	\$10.45	33.20%
Q4 High	2,629	\$8.20	32.00%

High-Sales vs. Low-Sales Vendors

High-sales vendors averaged a **31.88%** weighted margin; low-sales vendors averaged **30.35%**. A vendor-level Welch's *t*-test found **no statistically significant difference** ($t = 0.89$, $p = 0.3779$, Cohen's $d = 0.23$). The small effect size should guide investigation rather than support a claim that the groups operate under distinct profitability models.

Analytical Correction on Record

The previous 72% bulk-savings statement has been retired. Cross-product averages cannot isolate a quantity discount from the effects of product mix, package size, brand positioning, or negotiated supplier terms. Future cost claims must be supported by matched-product, controlled analyses.

§ 8. Recommended 30/60/90-Day Action Plan

Timing	Owner	Action	Success Measure
0–30 days	Procurement	Review top-ten vendor continuity, document alternates and lead times.	Continuity coverage documented; concentration monitored.
0–30 days	Merchandising	Freeze or reduce replenishment for highest-value, low-sell-through brands.	Ending retail value declines without avoidable stockouts.
31–60 days	Category Mgmt	Pilot 10–15 promotion candidates with minimum-margin guardrails.	Incremental units and margin improve together vs. baseline.
31–60 days	Finance + Procurement	Review freight rate once per vendor as a share of spend.	Freight-to-spend variance explained; no double-counting.
61–90 days	Analytics	Re-score vendors and brands after interventions using the same methodology.	KPIs compared against the current baseline with variance explained.

Monthly Decision Scorecard

Domain	Metrics
Supplier	Procurement spend, top-10 concentration, continuity coverage
Commercial	Sales, estimated gross profit, weighted margin, cost coverage
Inventory	Ending retail value, sell-through rate, turnover, stockout exceptions
Experiments	Promotion lift, incremental margin dollars, post-test sell-through

Method and Limitations

Item	Treatment
Source data	MySQL sales, purchases, invoices, price references, and inventory snapshots.
Estimated COGS	Units sold multiplied by weighted purchase cost for the same vendor-brand.
Margin	Gross profit divided by cost-matched sales; row-level percentages are never averaged directly.
Inventory value	Ending on-hand units multiplied by retail price; not cash locked at cost.
Reconciliation	Exact at portfolio level; does not prove record-level transaction lineage.
Causality	All observed associations are descriptive; none are labelled as causal savings.
Decision use	All recommendations are measurable pilots, not guaranteed financial outcomes.

Notebook sequence. `01_data_quality_and_exploratory_analysis.ipynb` establishes the Data Analyst evidence base; `02_vendor_performance_business_analysis.ipynb` converts it into Business Analyst decisions, owners, and success measures.